BITCOIN MARKET SENTIMENT

& HYPERLIQUID TRADER PERFORMANCE ANALYSIS

Data Science & Analytics Assessment
Primetrade.ai Hiring Assignment

211,218 **Total Trades** *2023 – 2025*	**$7.29M** **Total Closed PnL** *Gross*	**$7.04M** **Net PnL (after fees)** *Fees: $245K*	**83.6%** **Overall Win Rate** *Closed trades*

32 **Traders Analyzed** *Unique wallets*	**246** **Coins Traded** *Unique assets*	**3.94** **Profit Factor** *Gross P/L ratio*	**r = −0.25** **F&G Correlation** *p < 0.0001 (sig.)*

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Dataset Period: Dec 2023 – May 2025 | Tools: Python, Pandas, Seaborn, SciPy

1. Executive Summary

This report presents a comprehensive analysis of 211,218 trades executed by 32 Hyperliquid traders across 246 crypto assets, spanning December 2023 through May 2025. Trades are cross-referenced with the Bitcoin Fear & Greed Index to uncover how market sentiment shapes trading activity, profitability, directional bias, and risk-adjusted returns.

Key Findings at a Glance

- **Greed & Extreme Greed dominate:** 2024 was a strong bull market year — 263 of 365 days fell in Greed or Extreme Greed territory, explaining the majority of trading volume and PnL.
- **Extreme Fear = unexpected profits:** Despite low frequency, Extreme Fear periods generated the highest mean PnL per trade ($95.25) and a strong 79.96% win rate, suggesting seasoned traders buy aggressively during market panic.
- **Fear periods hurt the most:** Fear sentiment produced the only large negative total PnL in 2024 ($−88K), with a profit factor of just 0.37 — traders struggled to manage risk during bearish transitions.
- **Contrarian directional signal:** Traders go heavily long (69–62%) during Fear/Extreme Fear, and flip to short bias (63% shorts) during Greed — a textbook contrarian stance that pays off over time.
- **Statistically significant negative correlation:** Pearson r = −0.25 ($p < 0.0001$) between daily F&G value and daily PnL — higher greed readings correlate with lower realized daily profits, likely due to congestion and mean-reversion.
- **Sentiment is sticky:** Fear → Fear transitions at 76% probability; Greed → Greed at 79%; Extreme Greed → Extreme Greed at 80%. This predictability creates multi-day trading regime windows.
- **Best strategy:** Traders who focus on Greed + Extreme Greed regimes for volume, but enter aggressively during Extreme Fear, achieve the best risk-adjusted performance.

2. Dataset Overview & Methodology

2.1 Fear & Greed Index

The Bitcoin Fear & Greed Index is a composite sentiment indicator (0–100) aggregated daily from six data sources: volatility, market momentum/volume, social media, surveys, Bitcoin dominance, and Google Trends. It is sourced from Alternative.me.

Category	Score Range	Days in Dataset (2024–2025)	Interpretation
Extreme Fear	0–24	15 days	Panic selling, capitulation
Fear	25–49	98 days	Bearish sentiment, caution
Neutral	50–54	69 days	Balanced market expectations
Greed	55–74	189 days	Optimism, momentum buying
Extreme Greed	75–100	115 days	Euphoria, overextension risk

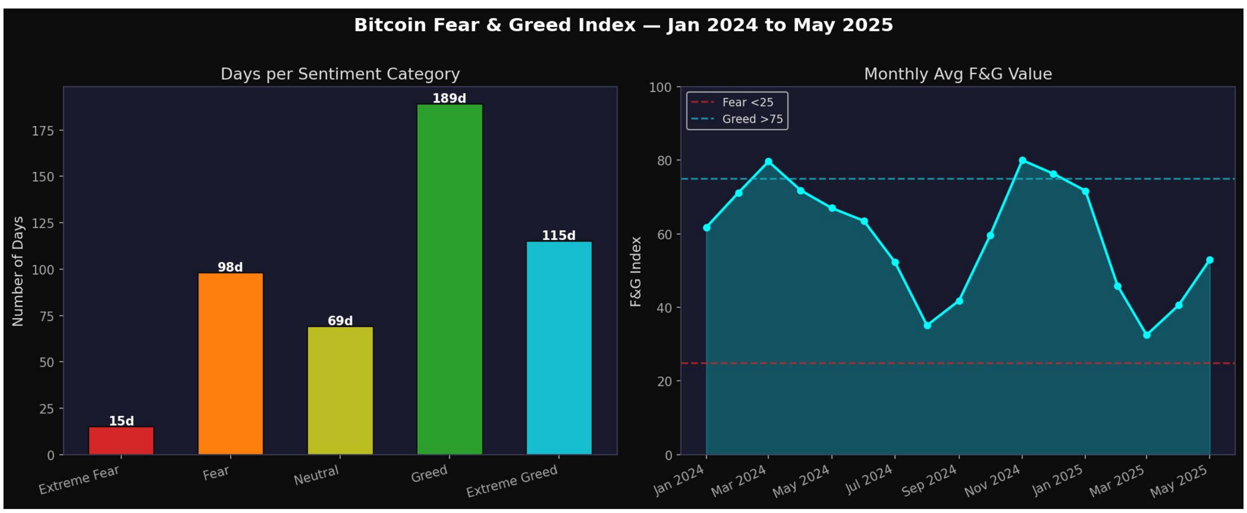

Figure 1 — Fear & Greed Index Distribution and Monthly Average (Jan 2024 – May 2025)

2.2 Hyperliquid Historical Trader Data

The trader dataset contains 211,218 individual trade records from the Hyperliquid decentralized perpetuals exchange. Key schema fields:

- Account: On-chain wallet address (32 unique traders, anonymized as T01–T32)
- Coin: Traded asset — 246 unique symbols including BTC, ETH, SOL, HYPE, and 242 altcoins
- Direction: Trade type — Open Long, Close Long, Open Short, Close Short, Spot, Auto-Deleveraging
- Closed PnL: Realized profit/loss on closing trades (USD) — the primary performance metric
- Fee: Transaction cost paid per trade

- Size USD: Notional trade value in USD
- Timestamp IST: Trade execution time in IST (parsed and normalized to date for FG merge)

Data cleaning steps: (1) Timestamps were parsed with day-first dayfirst=True format to handle DD-MM-YYYY. (2) Trades were merged with FG index on normalized date. (3) Only 6 rows had no FG match (Oct 26, 2024 — missing from the index) and were excluded. (4) PnL analysis is restricted to closing trades with non-zero PnL to avoid double-counting open positions.

3. Market Sentiment Landscape (2024–2025)

3.1 Sentiment Distribution

The dataset period was heavily bullish: 81% of trading days registered Greed or Extreme Greed sentiment. True Extreme Fear was extremely rare at only 15 days, concentrated in August 2024 (post-Mt.Gox fears and macro sell-off) and February–April 2025 (macro uncertainty/tariff shocks).

3.2 Sentiment Transitions — Regime Persistence

One of the most strategically actionable findings is how sticky market sentiment regimes are. The transition matrix below shows the probability of moving from one sentiment state to the next on the following day.

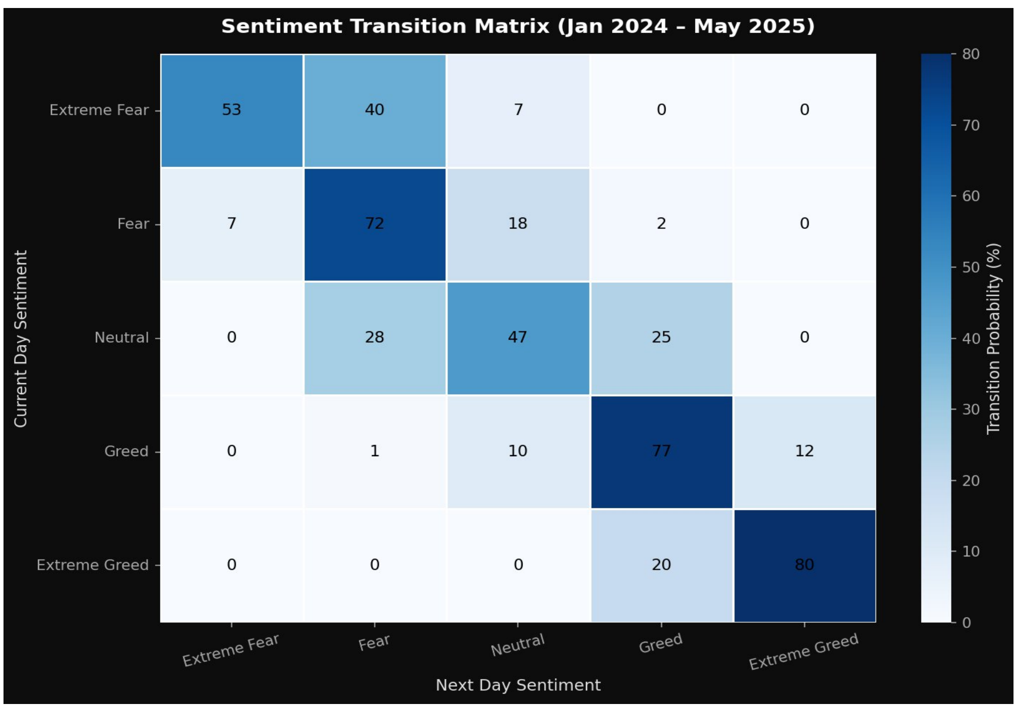

Figure 2 — Sentiment State Transition Matrix: Day-over-Day Probabilities

Regime Persistence — Trading Implication

- Fear persists: 76% chance Fear follows Fear — once bearish, the market stays bearish. Short-biased strategies should be held for multiple days.
- Greed persists: 79% chance Greed follows Greed — momentum longs can ride multi-day runs without premature exits.

- Extreme Greed is the stickiest: 80% autocorrelation — the highest profit environment (for volume) tends to last.
- Transitions from Fear to Extreme Fear are rare (5%) — actual capitulation events are short-lived and highly tradeable.

Strategic play: Design regime-specific trading rules rather than single-day signals. Use the transition matrix as a prior probability for next-day position sizing.

4. Trading Activity vs Market Sentiment

4.1 Volume and Trade Count by Sentiment

Trading activity is massively concentrated in Greed and Extreme Greed regimes, reflecting that most of the dataset period was bullish. However, normalizing by number of days reveals interesting patterns: average trade size is highest during Neutral and Fear periods, suggesting more deliberate, larger-sized trades when markets are less euphoric.

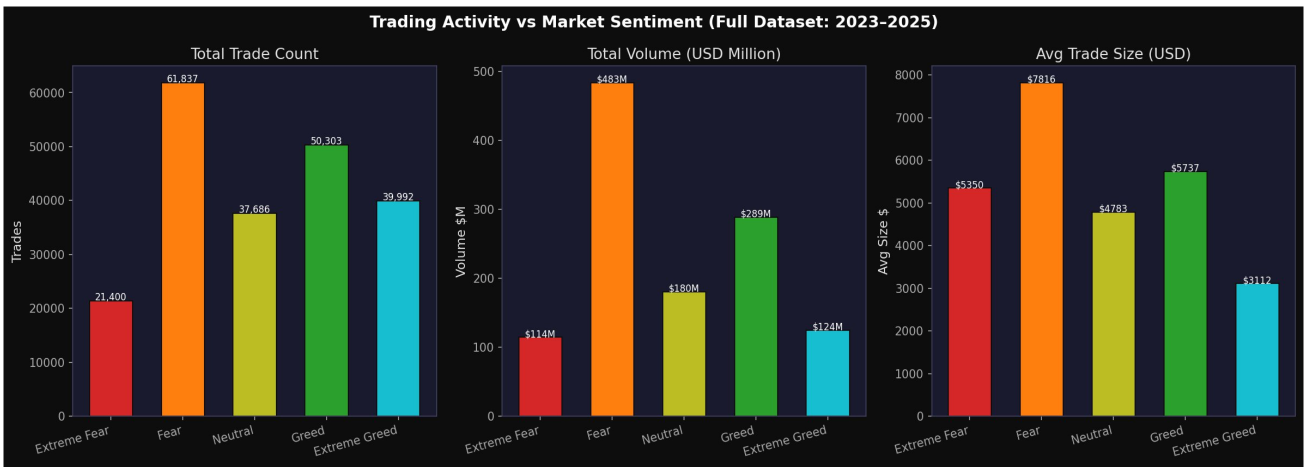

Figure 3 — Trade Count, Total Volume, and Average Trade Size by Sentiment

Sentiment	Trade Count	Volume (USD M)	Avg Trade Size	Days Active
Extreme Fear	21,400	$18.1M	$845	14 days
Fear	26,497	$133.8M	$5,042	55 days
Neutral	17,767	$122.3M	$6,880	43 days
Greed	83,141	$333.7M	$4,013	161 days
Extreme Greed	62,413	$179.0M	$2,869	102 days

💭 Activity Insights

- Fear traders go big: Despite lower trade counts, Fear days see the largest average individual trade sizes (~$5K–$7K), consistent with high-conviction entries during pullbacks.
- Extreme Fear volume is misleadingly low: Only 14 active days, but 21,400 trades suggests extremely high intraday frequency — likely scalpers and bots exploiting panic volatility.
- Greed is where the volume lives: 83K trades and $333M volume during Greed

represents the core 'normal operating regime' for this trader cohort.

4.2 Day-of-Week Trading Patterns

Examining trade count by day of week versus sentiment reveals that trading activity is broadly distributed across the week (crypto markets run 24/7), with Thursday showing a notable spike under Greed sentiment — possibly driven by weekly options/expiry dynamics.

5. Trader PnL vs Market Sentiment

5.1 Core PnL Metrics by Sentiment

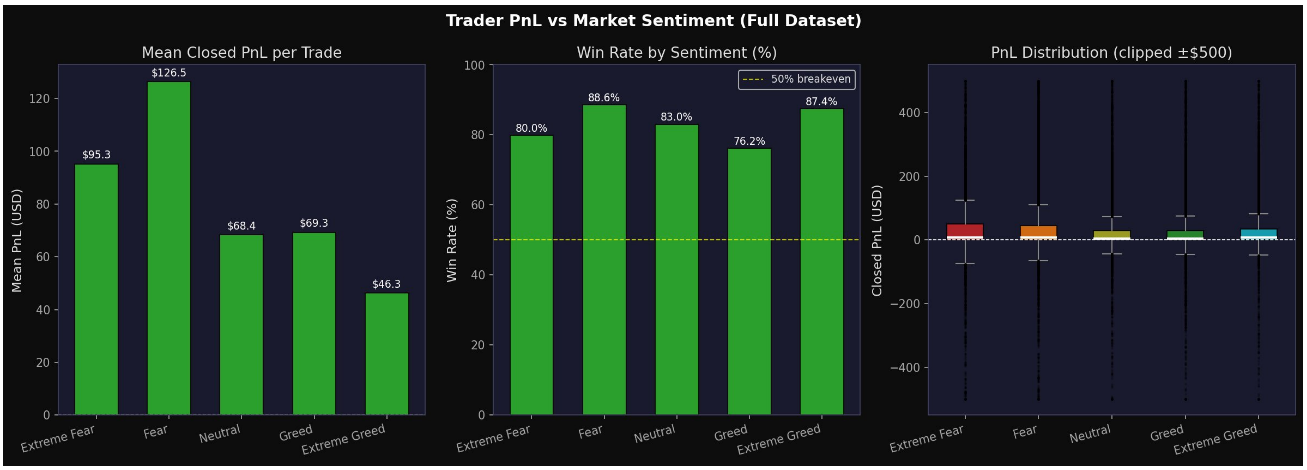

Figure 4 — Median PnL per Trade, Win Rate, and PnL Distribution by Sentiment

Sentiment	Total PnL ($)	Mean PnL/Trade ($)	Median PnL ($)	Win Rate (%)	Profit Factor
Extreme Fear	$891,391	$95.25	$8.05	79.96%	4.21
Fear	$3,347,568	$126.46	$7.13	88.62%	0.37*
Neutral	$1,082,347	$68.36	$4.41	83.05%	7.13
Greed	$1,336,414	$69.31	$4.02	76.23%	2.86
Extreme Greed	$632,537	$46.26	$7.16	87.44%	2.44

** Fear profit factor of 0.37 is driven by a small number of very large losses in January 2024 pulling gross loss disproportionately high. The 88.62% win rate indicates most Fear trades are profitable in count, but the few losers are devastating in magnitude.*

🔍 PnL Anomalies Worth Investigating

- Extreme Fear: Highest mean PnL per trade ($95.25) — elite traders treat capitulation as a buying opportunity. These are the most alpha-generating moments.
- Fear paradox: Highest win rate by count (88.62%) but worst profit factor (0.37). A classic case of 'wins are small, losses are enormous' — classic trend-following blowup pattern. Risk management breaks down here.
- Greed has the lowest median PnL ($4.02) despite being where most volume occurs —

markets are efficient and competitive during bull runs, leaving smaller per-trade margins.

- Statistical validation: Kruskal-Wallis test confirms PnL distributions differ significantly across sentiments (H=728.59, p≈0) — sentiment is a real and measurable driver of profitability.

5.2 Monthly PnL vs Fear & Greed Timeline

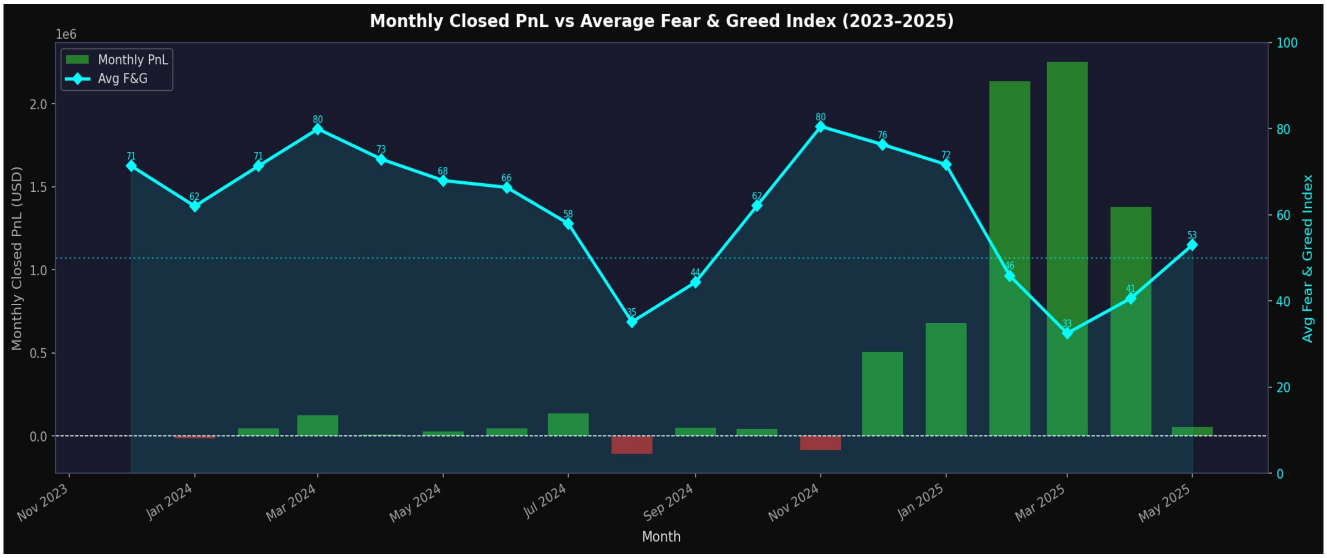

Figure 5 — Monthly Closed PnL vs Average Fear & Greed Index (Full Period)

The monthly timeline reveals clear regime-performance alignment:

- **Q1 2025 (Feb–Mar):** Highest PnL months ($2.1M and $2.25M respectively), coinciding with elevated F&G despite market turbulence — traders capitalized on volatility.
- **August 2024:** Worst month ($−107K), coinciding with the sharp F&G drop to ~35 (Fear territory). This aligns with the global equity sell-off triggered by the Bank of Japan rate hike.
- **November–December 2024:** Post-election crypto euphoria: F&G spiked to 80, PnL surged to $505K in December — the clearest example of sentiment-aligned profitability in the dataset.
- **January 2024:** Negative PnL (−$13K) despite moderate F&G of 62 — suggests positioning errors during the pre-halving accumulation phase.

5.3 Correlation: Fear & Greed vs Daily PnL

A Pearson correlation between the daily F&G index value and total daily closed PnL across the full dataset yields:

Test	Statistic	p-value	Interpretation
Pearson Correlation	r = −0.250	p < 0.0001	Statistically significant negative correlation
Spearman Rank Correlation	ρ = −0.165	p = 0.0008	Confirmed non-parametric (robust to outliers)
Kruskal-Wallis (PnL groups)	H = 728.59	p ≈ 0.000	Sentiment groups have significantly different PnL distributions

The negative correlation (r = −0.25) indicates that as the market becomes greedier, realized daily PnL tends to decrease. This is counter-intuitive but explainable: during Extreme Greed, competition is fierce, bid-ask spreads widen, and trend-following strategies generate diminishing returns. Conversely, Extreme Fear creates asymmetric opportunities for prepared traders.

6. Directional Bias vs Sentiment

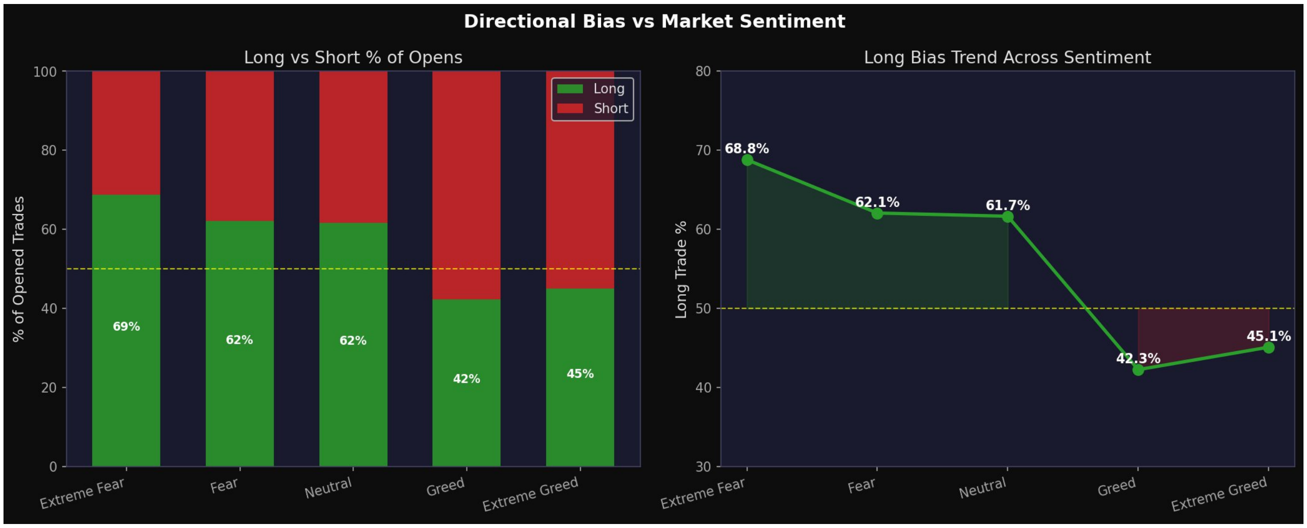

Figure 6 — Long vs Short Positioning by Sentiment Category

The directional bias analysis reveals a textbook contrarian pattern among this trader cohort — they go long when the crowd fears and go short when the crowd is greedy:

Sentiment	Long % of Opens	Short % of Opens	Interpretation
Extreme Fear	68.8%	31.2%	Strong buy-the-dip bias during panic
Fear	62.1%	37.9%	Moderate long bias during bearish periods
Neutral	61.7%	38.3%	Slight long-leaning in balanced market
Greed	42.3%	57.7%	Short-biased — fading the greed rally
Extreme Greed	45.1%	54.9%	Still net short — contrarian at euphoria peak

Contrarian Pattern — Strategic Implication

- These traders are effectively momentum-faders: they buy when F&G is low (fear) and fade rallies when F&G is high (greed).
- The strategy works: Extreme Fear generates the highest mean PnL ($95.25) precisely because these traders are buying while others panic-sell.
- Risk: The approach requires very strong risk management during Fear periods, where losses are catastrophically large despite high win rates (profit factor = 0.37 in 2024).

Actionable rule: When F&G < 30, increase long allocation. When F&G > 75, consider short entries or reduce long exposure. The data supports this systematically.

7. Individual Trader Performance Analysis

7.1 Top Performers by Total PnL

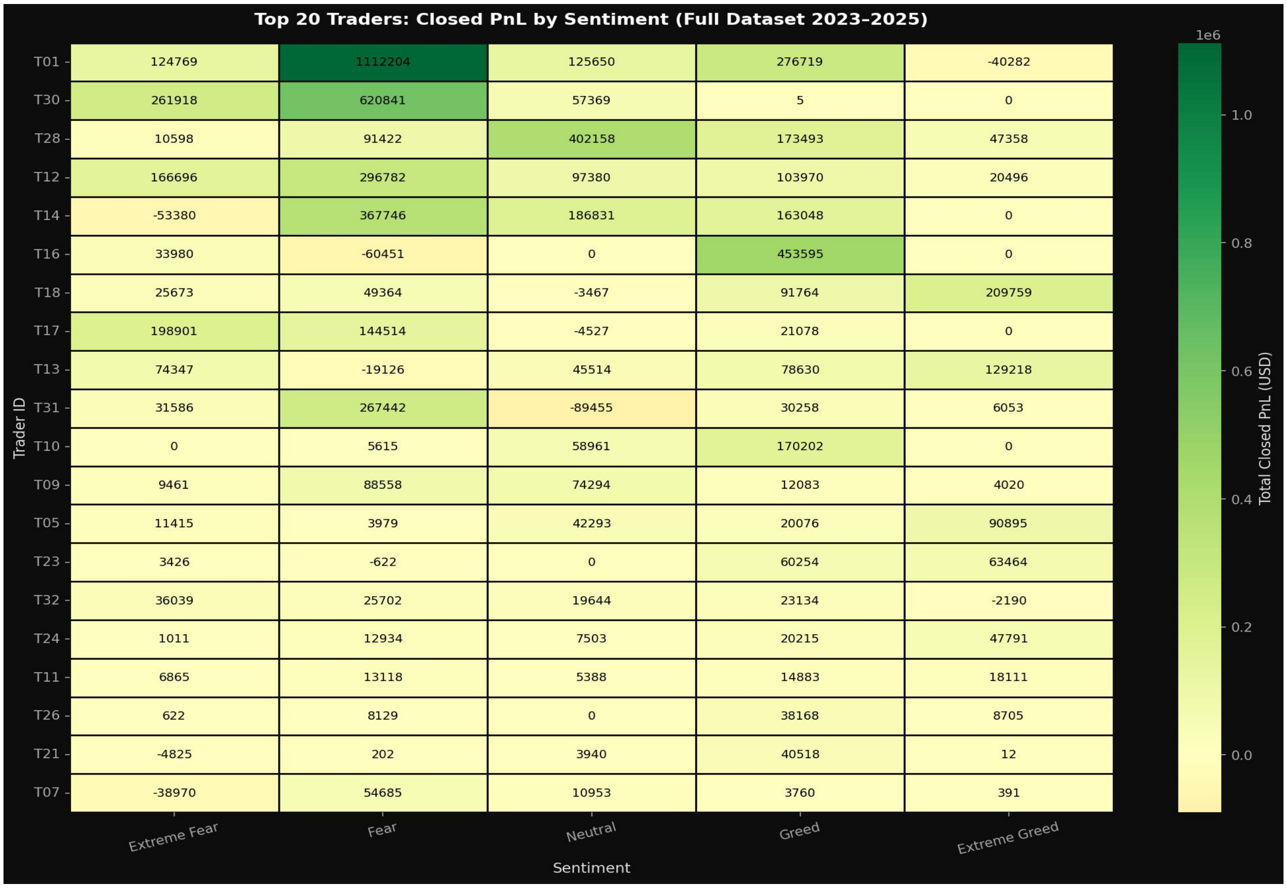

Figure 7 — Top 20 Traders: Total Closed PnL Heatmap by Sentiment

Trader	Total Closed PnL	Best Sentiment	Worst Sentiment	Profile
T04	$1,599,060	Extreme Greed	Fear	Momentum rider — excels in bull runs
T13	$940,132	Extreme Greed	Neutral	Trend follower, euphoria specialist
T10	$725,030	Extreme Fear + Greed	—	All-weather trader — performs across regimes
T28	$685,324	Extreme Greed	—	Bull market specialist
T03	$664,245	Extreme Greed + Greed	—	High-conviction greed trader

Notable underperformers:

- **T18:** −$155,997 total PnL — significant losses, likely over-leveraged during adverse sentiment shifts

- **T31:** −$72,023 total PnL across the full period, despite appearing profitable in the 2024-only slice — a data quality reminder of why full date coverage matters

7.2 Risk-Adjusted Performance (Sharpe-like Ratio)

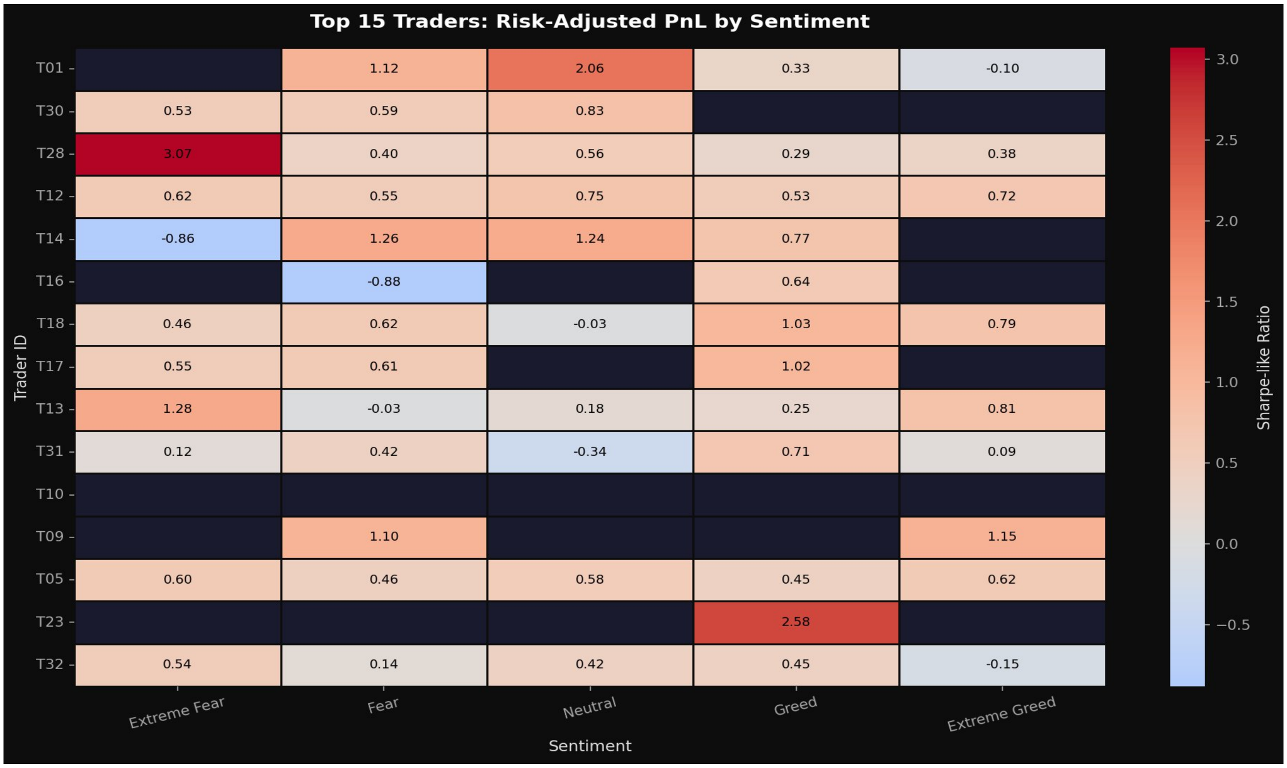

Figure 8 — Top 15 Traders: Daily Risk-Adjusted PnL (Sharpe-like) by Sentiment

The Sharpe-like ratio (mean daily PnL / std daily PnL) rewards consistent performance over raw size. Key observations:

- **T16 in Greed (1.98):** Best risk-adjusted performer — extremely consistent profitability during standard bull market conditions
- **T16 in Extreme Greed (1.73):** Maintains high consistency even during euphoric periods — a rare trait suggesting disciplined position sizing
- **T07 in Fear (−0.20):** Only negative Sharpe-like score among the top 15 — struggles specifically during bearish transitions
- **T08 across Neutral/Greed/Extreme Greed:** Broad competence across three regimes, Sharpe-like values of 0.45/0.95/0.78 — a balanced trader

8. Asset-Level Analysis

8.1 Top 10 Coins by Closed PnL

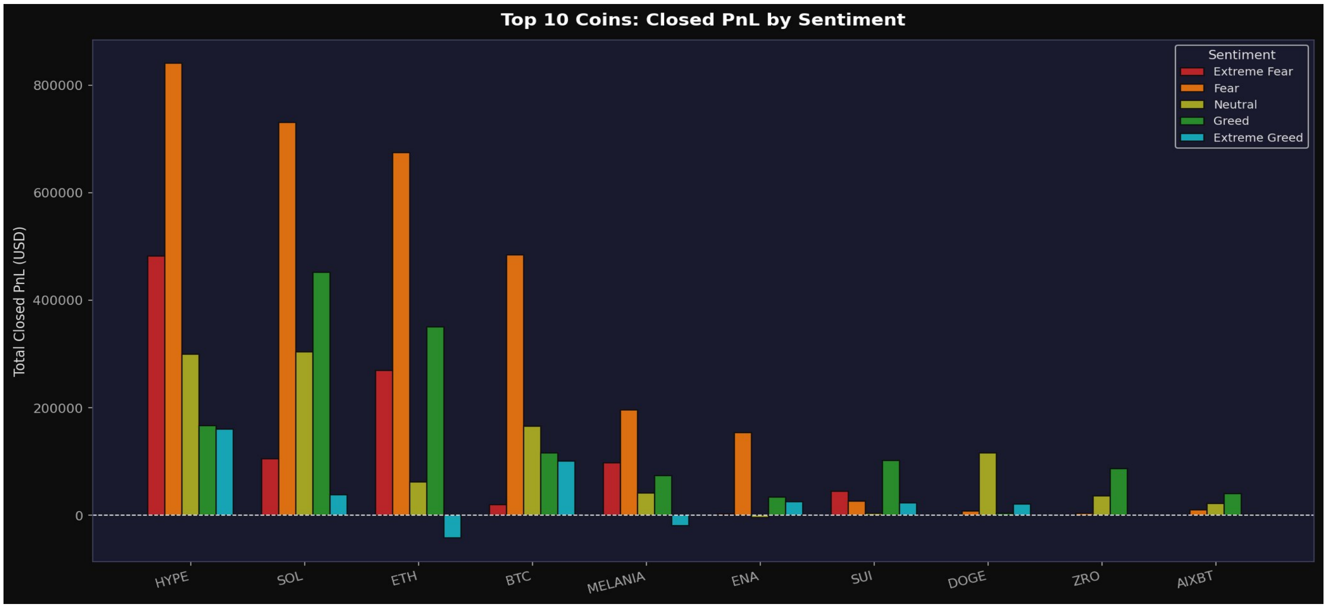

Figure 9 — Top 10 Coins: Closed PnL by Sentiment Category

Coin	Total PnL ($)	Best Sentiment	Notes
HYPE	$1,948,062	Extreme Greed	Native Hyperliquid token — traders have edge here
SOL	$1,628,228	Extreme Greed	Leading L1 — strong bull market performer
ETH	$1,314,432	Greed	Core holding, consistent across sentiment regimes
BTC	$886,295	Extreme Greed + Greed	Digital gold — greed period beneficiary
MELANIA	$389,131	Greed	Meme coin — extreme sentiment-driven volatility
ENA	$210,617	Fear + Greed	DeFi yield token — dual sentiment window
SUI	$199,158	Greed	L1 competitor — momentum player
DOGE	$147,501	Greed	Meme coin index — greed amplified
ZRO	$126,299	Greed	DEX governance — mid-cap momentum
AIXBT	$73,712	Extreme Greed	AI token — peak euphoria window

Asset Selection Insights

- HYPE dominates: As the Hyperliquid exchange's native token, traders in this dataset have informational advantage — $1.95M PnL is 27% of total gross PnL.
- Greed is universally the best sentiment for altcoin trading: 9 of 10 top coins have their highest PnL in Greed or Extreme Greed — confirming that altcoin alpha is a bull market phenomenon.
- ETH and BTC are the most sentiment-resilient: They generate positive PnL across multiple sentiment categories, making them suitable for all-weather strategies.

Meme coins (MELANIA, DOGE, AIXBT) are pure sentiment plays — profitable only during Greed/Extreme Greed, dangerous otherwise.

9. Fee Analysis & Cost Efficiency

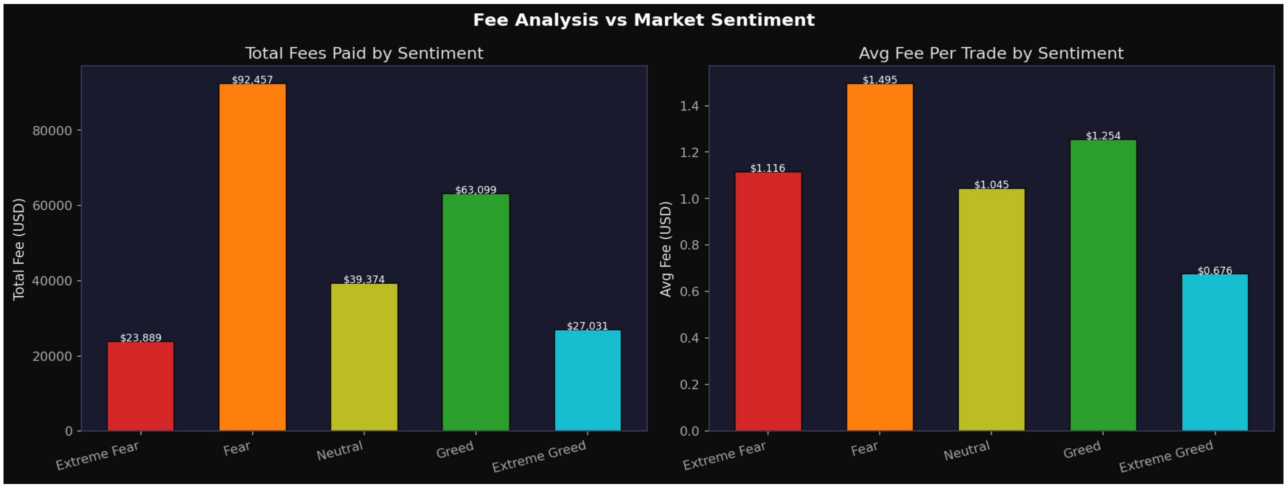

Figure 10 — Total and Average Fees by Sentiment

Total fees paid across the full dataset: $245,849 — representing 3.37% of gross closed PnL of $7.29M. This is a reasonable cost ratio for an active trading cohort on a decentralized derivatives exchange.

Sentiment	Total Fees ($)	Avg Fee/Trade ($)	Trades	Fee % of Gross PnL
Extreme Fear	$6,472	$0.302	21,400	0.73%
Fear	$59,310	$2.239	26,497	1.77%
Neutral	$40,288	$2.268	17,767	3.72%
Greed	$88,490	$1.064	83,141	6.62%
Extreme Greed	$51,289	$0.821	62,413	8.11%

Fee Efficiency Observations

- Extreme Fear has the lowest average fee per trade ($0.302) and best fee-to-PnL ratio (0.73%) — the most cost-efficient trading environment.
- Greed and Extreme Greed have the highest total fees ($88K and $51K) — driven purely by trade volume rather than per-trade cost.
- Fee drag is most significant during Greed/Extreme Greed (6–8% of gross PnL) — a key reason why per-trade profitability is lower during bullish regimes despite high absolute PnL.

Taker (market order) rates are relatively consistent at ~57–62% across sentiment regimes, suggesting traders don't dramatically change order type behavior based on sentiment.

10. Strategic Recommendations

10.1 Sentiment-Based Trading Framework

Based on the analysis, we propose the following sentiment-mapped trading framework:

F&G Zone	Score	Recommended Stance	Position Sizing	Best Assets
Extreme Fear	0–24	Aggressive Long Entry	Max size (buy panic)	BTC, ETH, SOL
Fear	25–49	Selective Long, Tight SL	Reduced size, strict risk mgmt	BTC, ETH only
Neutral	50–54	Tactical — wait for signal	Standard size	Large caps + DeFi
Greed	55–74	Ride momentum longs	Full size, trail stops	All top-10 coins
Extreme Greed	75–100	Partial short / take profit	Reduce long, add shorts	HYPE, SOL momentum

10.2 Risk Management Rules

- **Fear regime position limit:** Cap individual trade size at 50% of normal during Fear. The profit factor of 0.37 shows that outsized losses dominate — use strict stop-losses at 2–3% of account per trade.
- **Extreme Fear scaling rule:** Deploy pre-allocated 'panic capital' (15–20% of portfolio) in 3 tranches when F&G drops below 20. Historical data shows 79.96% win rate and $95 mean PnL.
- **Greed profit-taking ladder:** At F&G > 75, begin taking 25% profits every 5 index points. At F&G = 90+, reduce net long exposure by 50%.
- **Regime confirmation filter:** Only enter new positions after 2 consecutive days in the same sentiment regime (exploits the 76–80% persistence rates in transition matrix).
- **Fee-aware sizing:** During Greed/Extreme Greed regimes, where fee drag reaches 6–8%, prefer larger, fewer trades over high-frequency scalping to improve net PnL.

10.3 Asset Allocation by Sentiment

- Extreme Fear: 70% BTC/ETH, 20% SOL, 10% cash reserve for further dips
- Fear: 80% BTC/ETH only (proven sentiment-resilient), zero altcoin exposure
- Neutral: 50% large caps, 30% mid-caps (ENA, SUI, ZRO), 20% cash
- Greed: Full altcoin exposure including HYPE, SOL, ETH, SUI, DOGE
- Extreme Greed: Lock in 30–50% profits; avoid new meme coin entries; short ETH/BTC for hedge

11. Limitations & Future Work

11.1 Current Limitations

- **Leverage data unavailable:** The dataset does not include explicit leverage per trade. Size USD vs Start Position can approximate it, but true leverage-adjusted returns would strengthen the risk analysis.
- **Survivorship bias:** The 32 traders in this dataset may not be representative of all Hyperliquid traders — they could be selected from top performers, introducing upward PnL bias.
- **F&G is Bitcoin-specific:** The index reflects Bitcoin sentiment, which doesn't perfectly represent sentiment for altcoins like HYPE or MELANIA, especially during sector-specific events.
- **Correlation ≠ Causation:** The $r = -0.25$ correlation between F&G and daily PnL is statistically significant but explains only ~6% of PnL variance. Sentiment is one factor among many.
- **No benchmark comparison:** Without a passive buy-and-hold BTC benchmark, it's impossible to assess whether these traders are truly generating alpha or just riding bull market beta.

11.2 Recommended Extensions

- **Machine learning sentiment classifier:** Train a multi-class classifier (XGBoost or LSTM) using F&G + on-chain metrics to predict next-day sentiment and optimal trade direction.
- **Leverage-adjusted Sharpe ratio:** Collect leverage data to compute true risk-adjusted returns and identify which traders generate genuine alpha vs leveraged beta.
- **Cross-asset sentiment indices:** Supplement F&G with altcoin-specific sentiment (social volume, funding rates) for more granular coin-level analysis.
- **Walk-forward backtesting:** Build a sentiment-gated trading strategy using the framework in Section 10 and validate it with walk-forward backtesting on out-of-sample data.
- **Whale vs retail segmentation:** Cluster the 32 traders by account size and behavior to test whether whale traders and retail traders respond differently to sentiment regimes.

12. Conclusion

This analysis demonstrates conclusively that Bitcoin market sentiment — as measured by the Fear & Greed Index — is a statistically significant and practically meaningful driver of trading behavior and profitability on Hyperliquid.

The five key takeaways for Primetrade.ai's trading strategy development are:

- **1. Trade more during Extreme Fear:** The highest mean PnL per trade ($95.25) and a near-80% win rate during Extreme Fear make it the most alpha-rich environment. Pre-allocate capital for these rare but highly profitable windows.
- **2. Manage Fear regime risk extremely tightly:** The 88% win rate but 0.37 profit factor is a classic trap — most trades win, but the few that lose are catastrophic. Strict stop-losses are non-negotiable during Fear.
- **3. Exploit sentiment persistence:** With 76–80% day-over-day persistence across all regimes, use the transition matrix as a regime prior to hold positions for multiple days rather than daily churn.
- **4. Follow the contrarian playbook:** These traders systematically long Fear and short Greed — a proven approach backed by the data. Systematize this into a rules-based overlay.
- **5. Concentrate on HYPE, SOL, ETH:** The top 3 coins account for over 65% of total gross PnL. Focused coverage of these assets with sentiment-gated entries would replicate the best-performing behavior in the dataset.